ELEC70074
EE9-CS6-2
EE9-FPN1-03

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

POWER SYSTEM DYNAMICS STABILITY AND CONTROL

Friday, 16 May 2025, 10:00 AM

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : Y. Gu

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The Questions

1.

a)

i) List one dynamic event with its defining characteristic that can be classed as wave phenomenon in power systems.

[2]

ii) Briefly explain the influence of this dynamic event on the stability/control/protection design considerations in power systems.

[4]

b) Normally stator transients are neglected in transient or first rotor swing stability:

i) Why is it so?

[2]

ii) How will the first swing stability qualitatively/quantitively be impacted when these are included ?

[3]

iii) How are these stator transients modelled for electromechanical stability studies involving torsional stability in multi-stage turbine mass ?

[2]

iv) Why can not these transients be ignored for stability analysis of Type 3 /Type 4 wind turbines?

[3]

c)

i) How do the laminated stator and solid rotor of a synchronous machine help power system stability ?

[4]

ii) Why do the reactances of stator windings drop suddenly when they start feeding fault currents ?

[3]

iii) Why is it not advisable to operate a synchronous generator in leading power factor (VAR absorption) mode from a stability perspective?

[2]

[25 marks]

2.

a)

i) What is non-minimal phase transfer function in the contexts of hydro turbines?

[3]

ii) How does it impact the stability of speed governing for frequency response ?

[3]

iii) What control strategies are adopted to counter any adverse impact of the non-minimal phase transfer function behaviour?

[2]

b) A round rotor star connected synchronous generator is connected to the grid at 15.75 kV (line to line) voltage. The rated power is 210 MW, power factor is 0.85 (lagging). The synchronous reactance is 1.7 pu on its rated base of 247 MVA. The resistance is negligible. Find out:

i) the rated phase current in kA

[2]

ii) the excitation phase voltage in kV

[3]

iii) load angle (δ) in degrees

[2]

iv) load angle stability margin in degrees

[1]

v) total terminal reactive power in MVAR

[3]

c) The generator from part (b) is still considered. The excitation voltage is reduced by 10% while mechanical input remains unchanged.

i) Find out the change in steady state stability margin in degree with respect to the rated excitation voltage.

[3]

ii) Find out the change in total terminal reactive power with respect to operation at the rated excitation voltage.

[3]

[25 marks]

3.

a)

i) Why are variables (current,voltage , flux etc.) in dq (Park reference frame) needed to be transformed further through network (Kron reference frame) for multi-machine dynamic analysis ?

[3]

ii) Establish the relationship between the voltage between the two reference frames for machine load angle δ and machine terminal voltage (V_t) angle θ when they are with respect to the network reference frame.

[5]

iii) For $\delta = 30^0$ and $\theta = 5^0$, find out the d and q components of line terminal voltage of 20 kV in machine reference farme.

[3]

iv) What are the purposes amortissuer windings in synchronous machine and how are they modelled for stability studies?

[2+2]

b)

i) Why is neither caged rotor (Type 1) nor wound rotor (Type 2) wind turbine generator (WTG) the present preferred choice of the industry?

[4]

ii) What is stall in respect of WTG operation ?

[3]

iii) `Some types of stall controls are in practice and some types are discontinued` – Justify this with reasons.

[3]

[25 marks]

4.

a)

i) What is the capacity factor of a wind turbine?

[2]

ii) Why do offshore wind turbines have larger capacity factor over their onshore counterpart?

[2]

iii) What is the tip-speed ratio of a wind turbine generator ?

[3]

iv) How does the tip-speed ratio help in optimal power control under varying wind speed ?

[3]

v) Why is the turbine side mechanical control adopted only above the rated wind speed ?

[2]

b)

i) What is the advantage of Type 4 WTGs over their Type 3 counter parts from the dynamics and control point of view

[4]

ii) How are pitch control WTGs modelled for stability studies ?

[2]

iii) `Within a wind farm AC network, the oscillatory stability is more important than transient stability' – Justify this statement with reasons.

[3]

iv) How is *wake* in wind farm modelled for stability studies ?

[2]

v) How does *wake* impact the frequency stability capability and delivery of a wind farm?

[2]

[25 marks]